Epithelial MP10 | Cohort and estimability

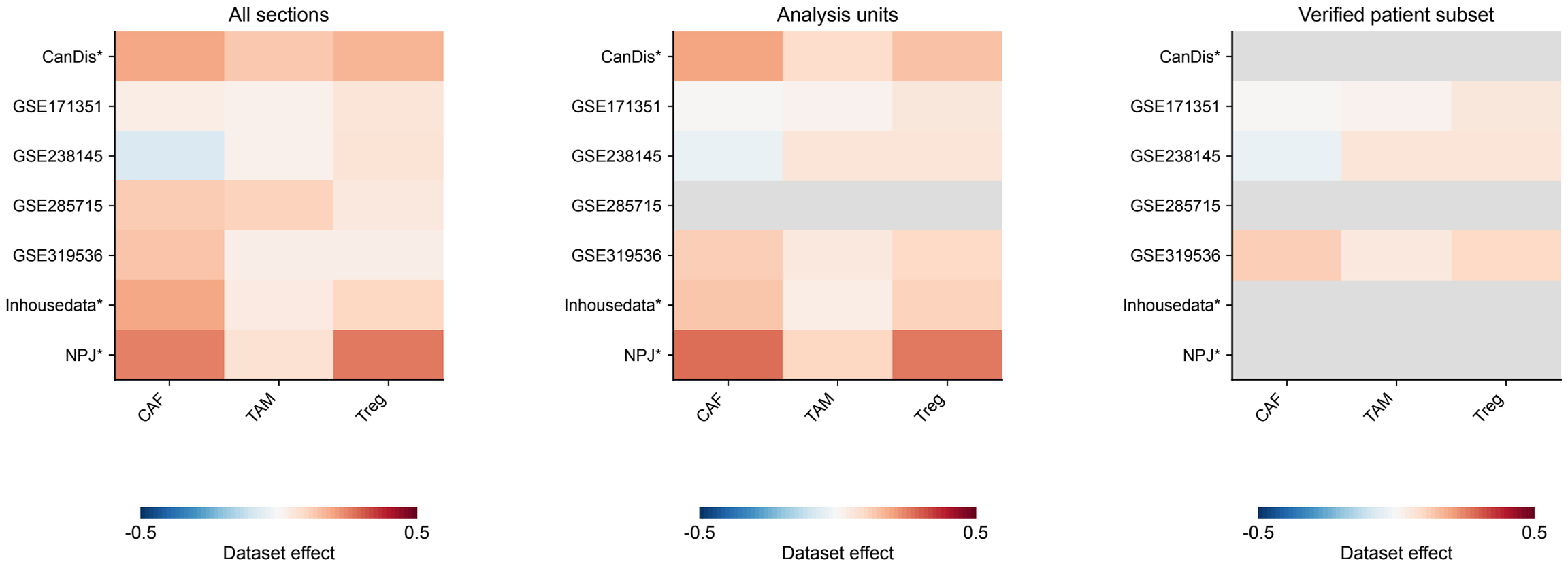
A section counts for a distance module when all three routes are estimable. Source groups are kept separate.

Dataset / source	sections	spots	queries	units	verified	P80	margin	adjusted	matched	UCell55
GSE285715 (normal)	3	5628	2309	0	0	3	0	3	3	3
CanDis*	13	56357	42244	13	0	13	12	13	13	13
GSE171351	4	4086	2915	4	4	3	3	3	3	4
GSE238145	10	74264	27122	9	9	9	9	9	8	10
GSE285715	3	11613	11305	0	0	3	3	3	3	3
GSE319536	16	58041	29323	16	16	16	15	16	16	16
Inhousedata*	7	31213	11981	7	0	7	6	7	7	7
NPJ*	7	23800	16262	7	0	7	7	7	7	7

* Original dataset alias unverified. Units include eligible anonymous sample units; verified counts are policy-eligible patient IDs.

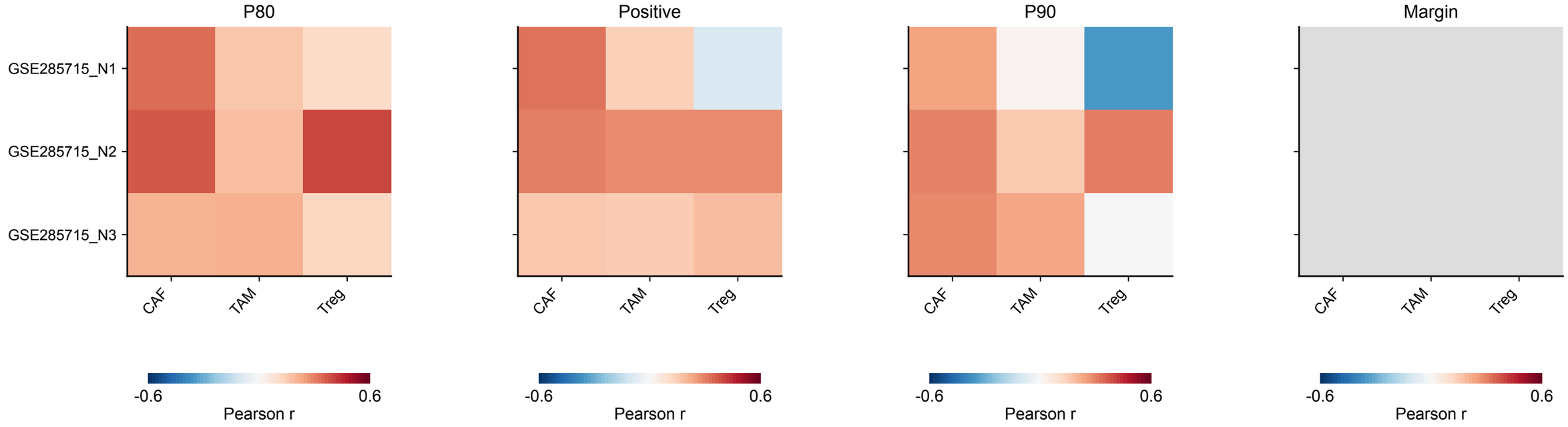
Tumor study sections | Main effects by dataset

Left: median section effect. Middle and right: equal-unit Fisher-z means within dataset. Grey indicates no eligible units.



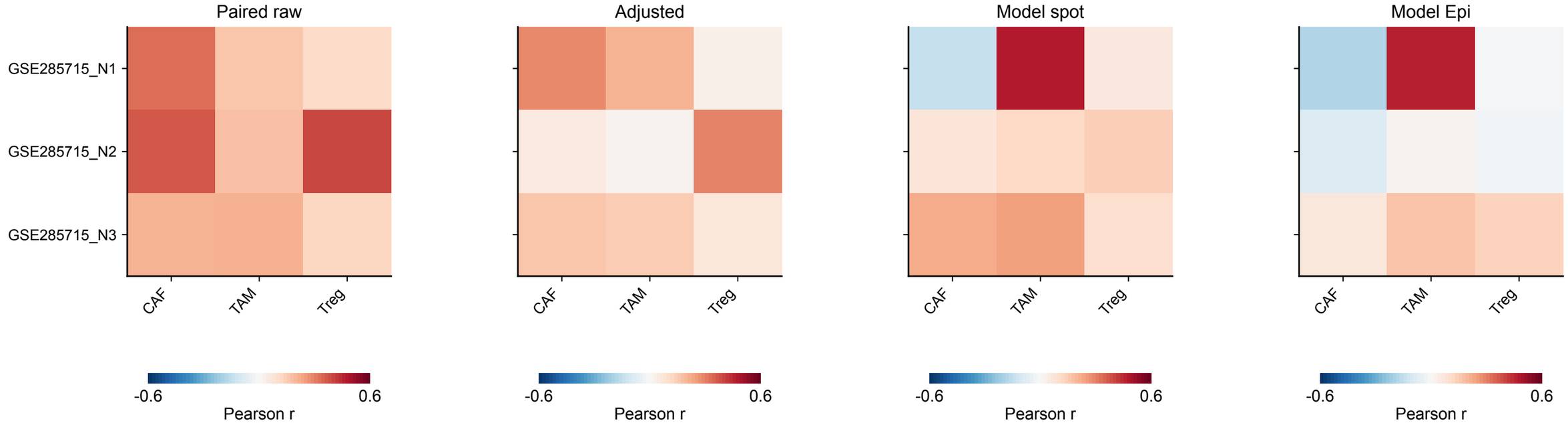
normal 1 | MP10–distance effects

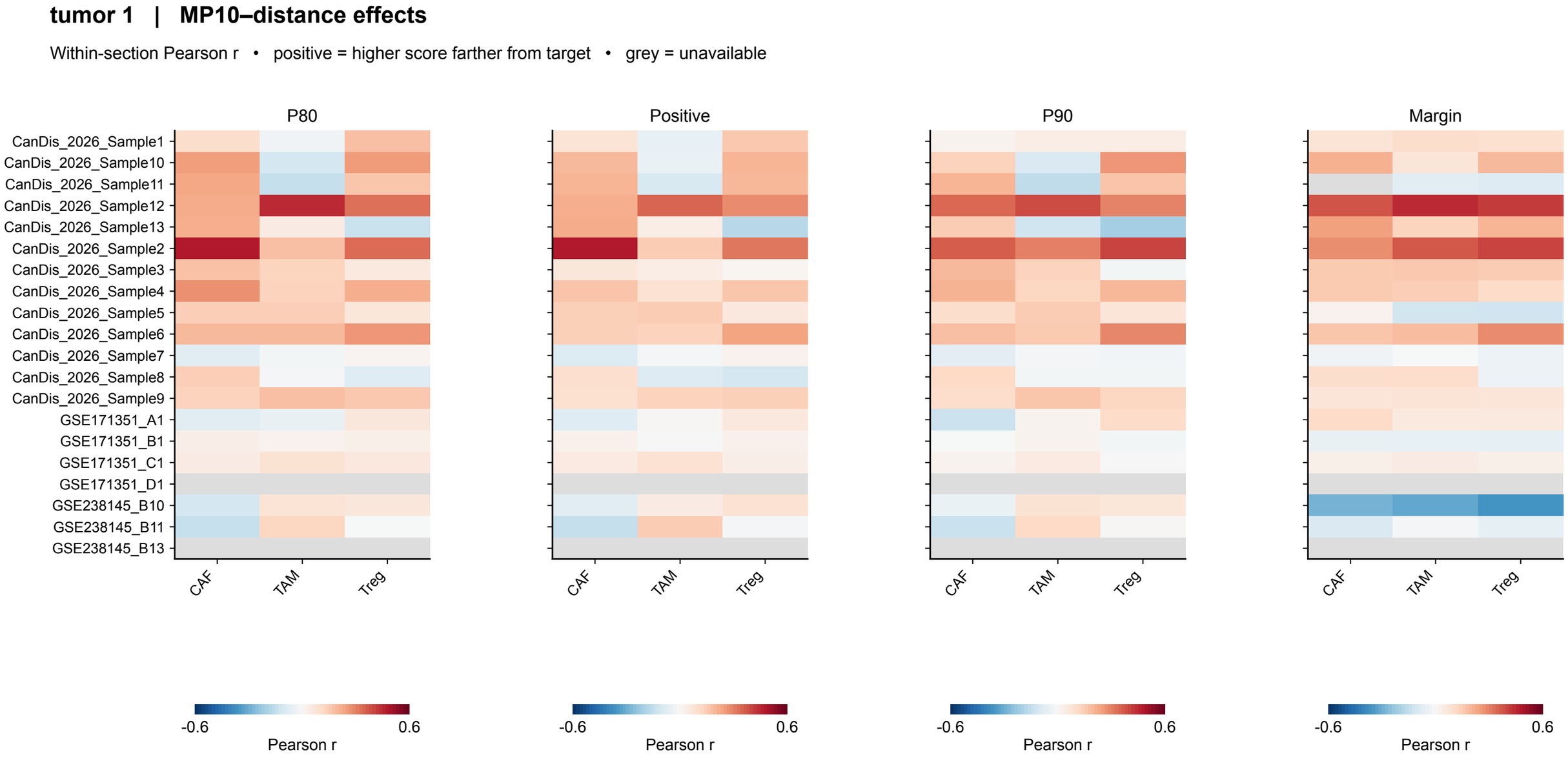
Within-section Pearson r • positive = higher score farther from target • grey = unavailable



normal 1 | MP10–distance effects

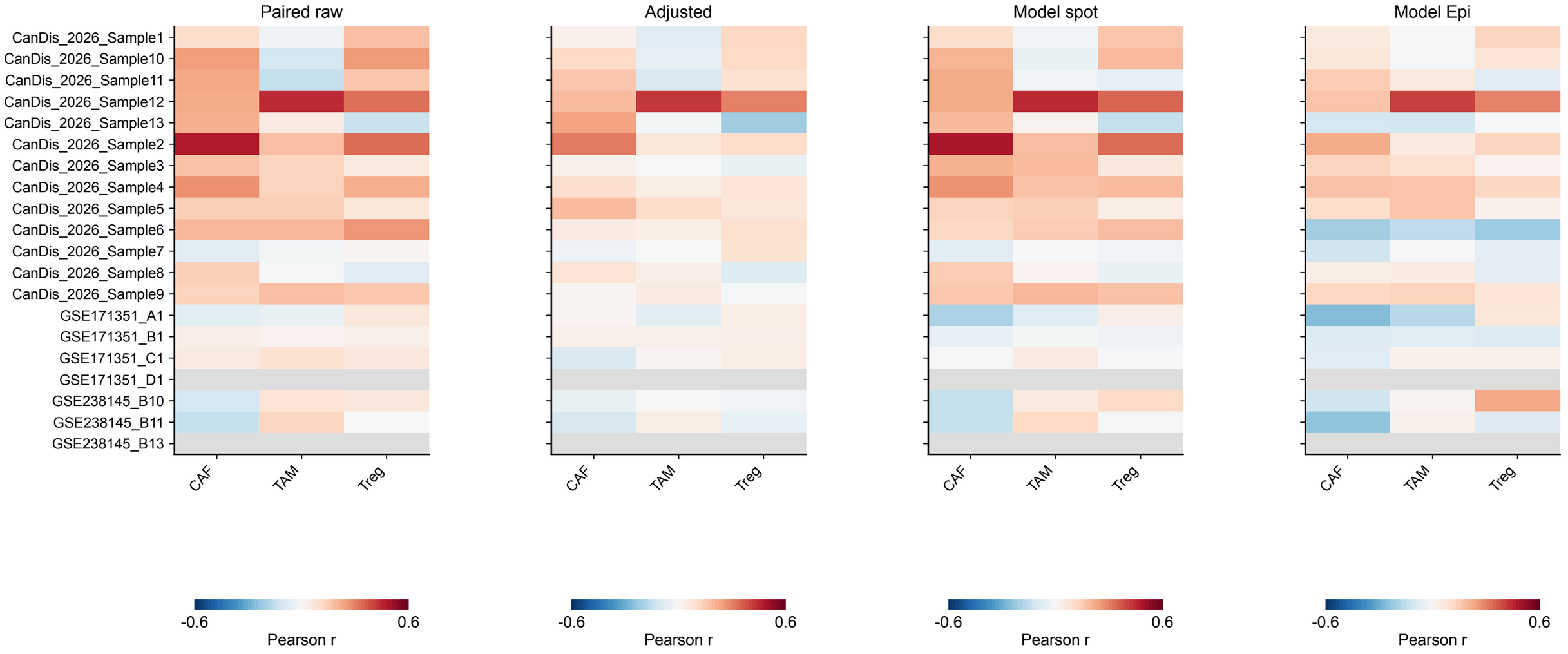
Within-section Pearson r • positive = higher score farther from target • grey = unavailable




tumor 1 | MP10–distance effects
Within-section Pearson r • positive = higher score farther from target • grey = unavailable
P80
Positive
P90
Margin
CanDis_2026_Sample1
CanDis_2026_Sample10
CanDis_2026_Sample11
CanDis_2026_Sample12
CanDis_2026_Sample13
CanDis_2026_Sample2
CanDis_2026_Sample3
CanDis_2026_Sample4
CanDis_2026_Sample5
CanDis_2026_Sample6
CanDis_2026_Sample7
CanDis_2026_Sample8
CanDis_2026_Sample9
GSE171351_A1
GSE171351_B1
GSE171351_C1
GSE171351_D1
GSE238145_B10
GSE238145_B11
GSE238145_B13
CAF
TAM
Treg
-0.6
0.6
Pearson r

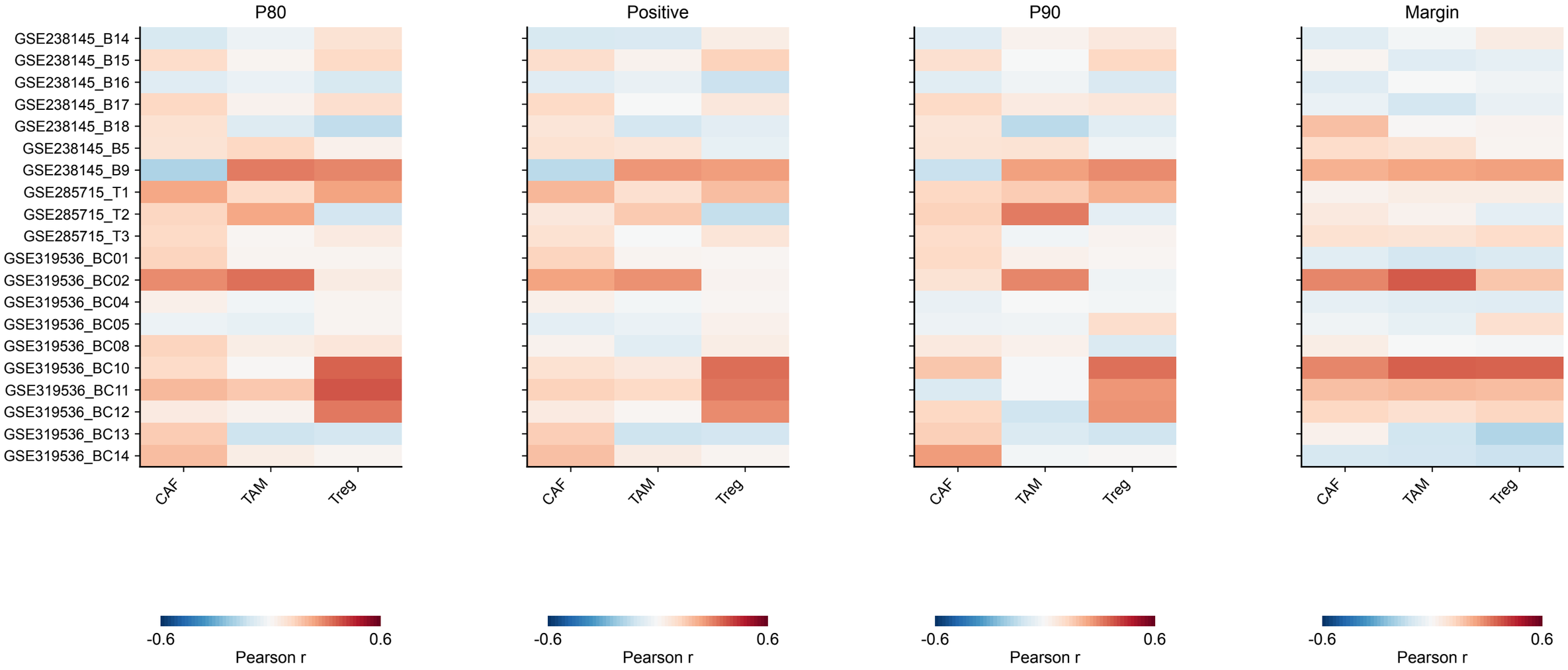
tumor 1 | MP10–distance effects

Within-section Pearson r • positive = higher score farther from target • grey = unavailable



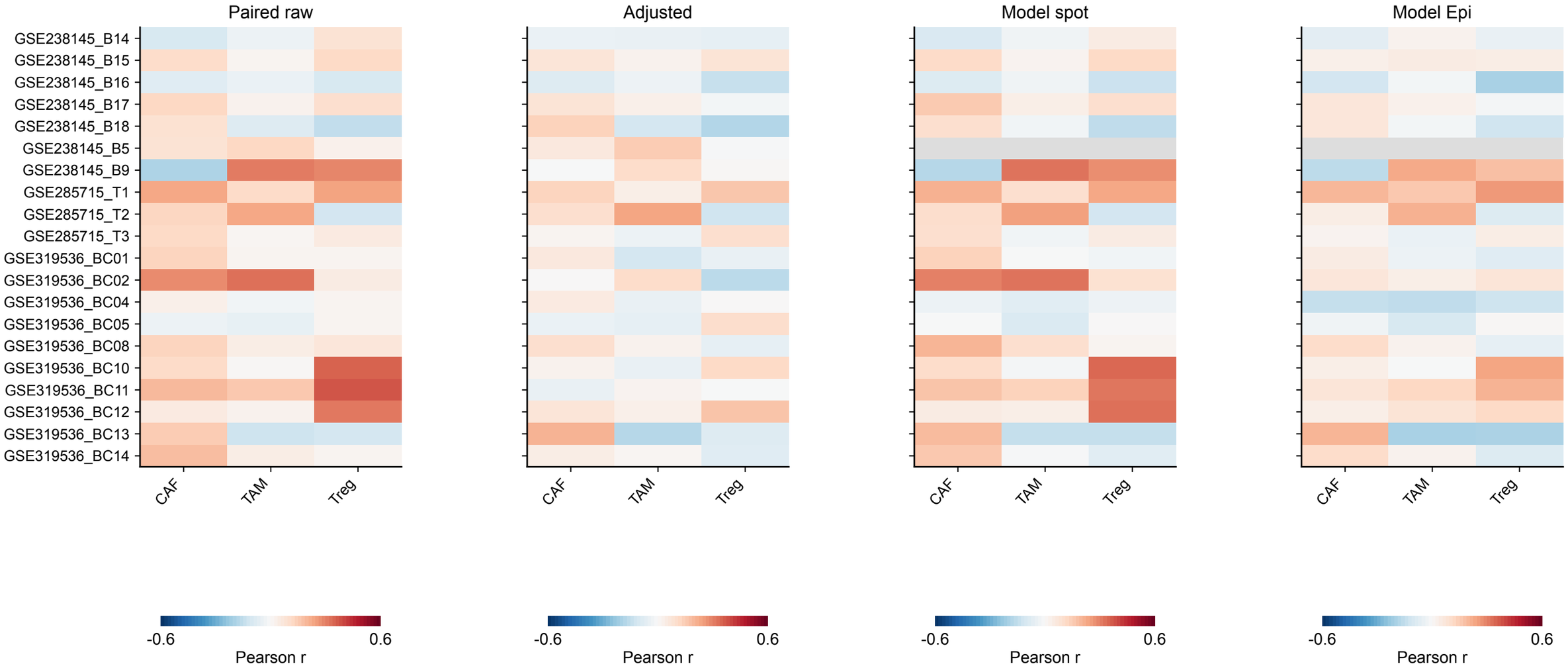
tumor 2 | MP10–distance effects

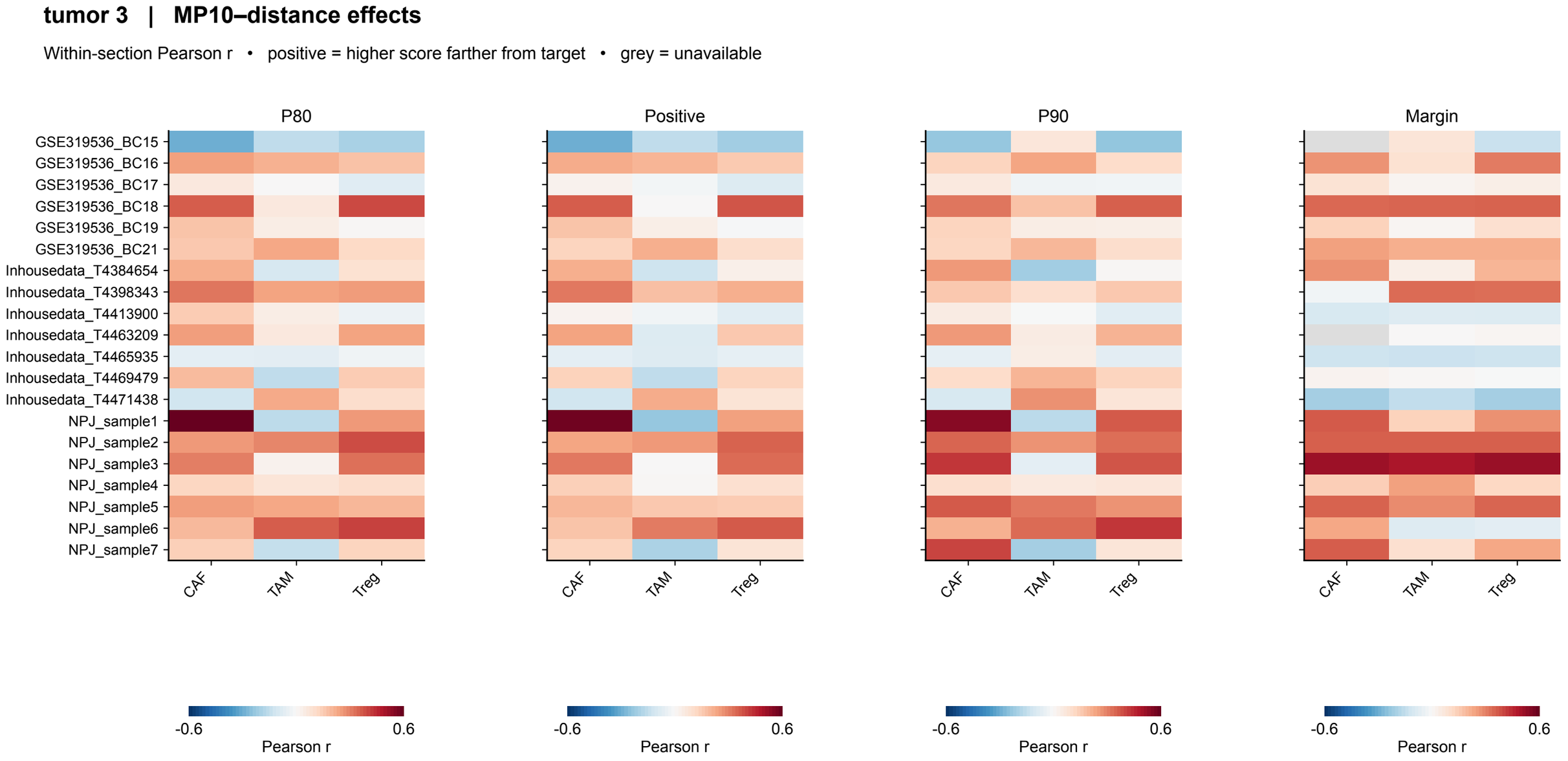
Within-section Pearson r • positive = higher score farther from target • grey = unavailable

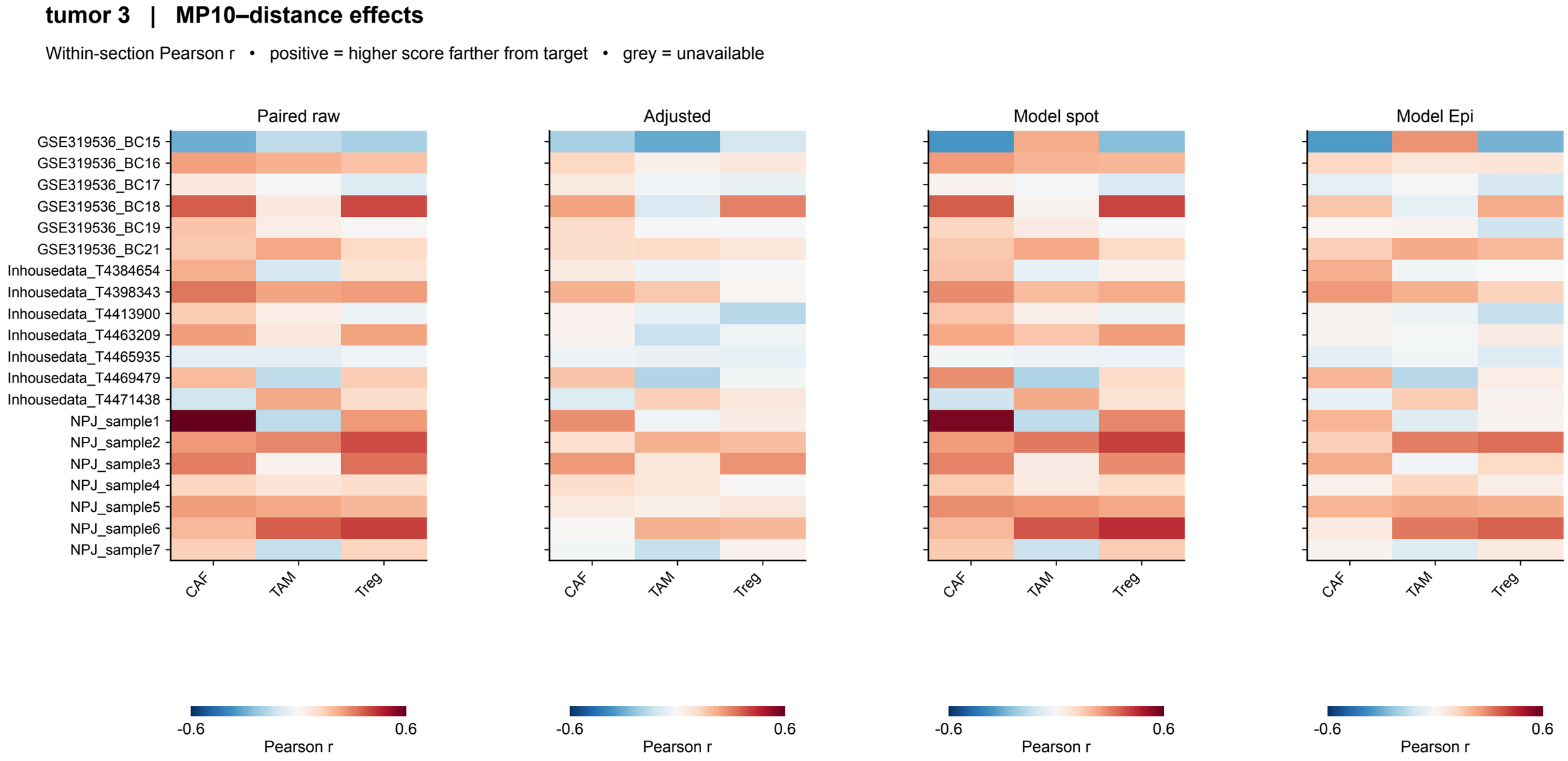


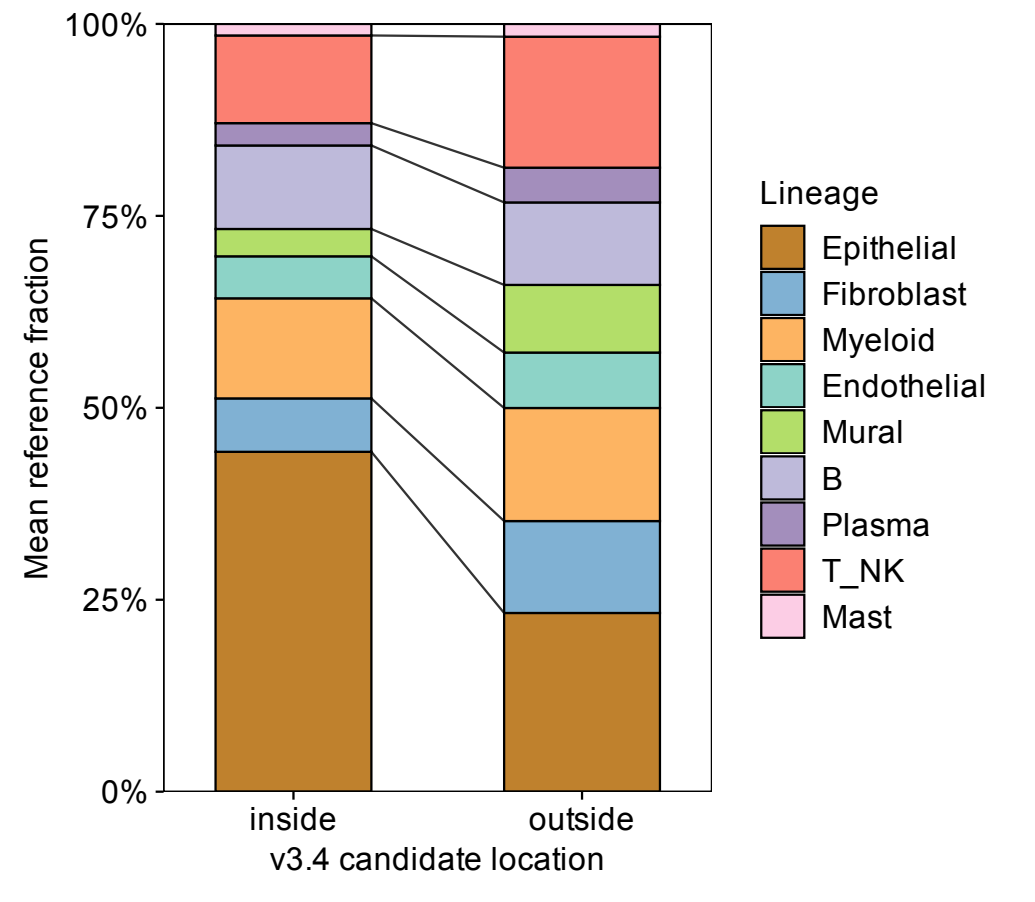
tumor 2 | MP10–distance effects

Within-section Pearson r • positive = higher score farther from target • grey = unavailable



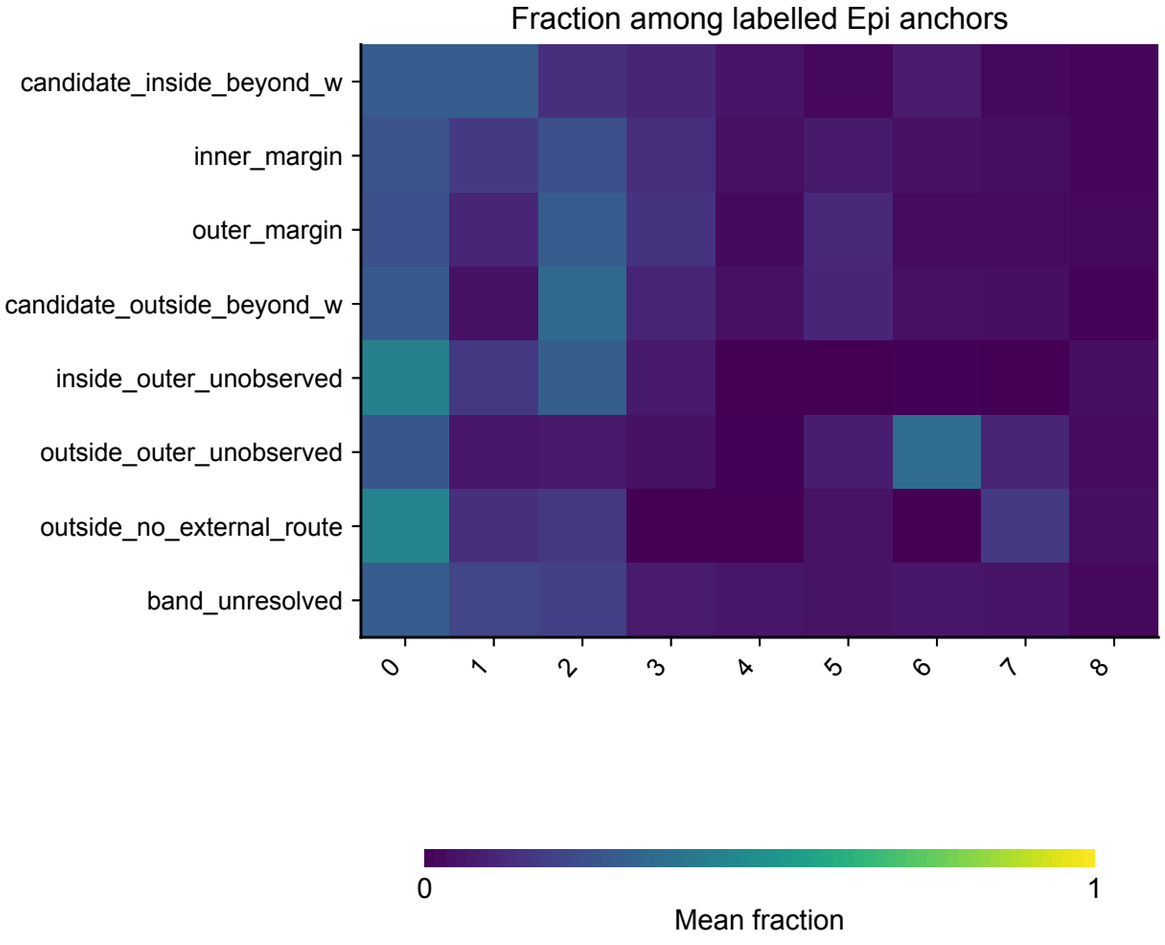

tumor 3 | MP10–distance effects
Within-section Pearson r • positive = higher score farther from target • grey = unavailable
P80
Positive
P90
Margin
GSE319536_BC15
GSE319536_BC16
GSE319536_BC17
GSE319536_BC18
GSE319536_BC19
GSE319536_BC21
Inhousedata_T4384654
Inhousedata_T4398343
Inhousedata_T4413900
Inhousedata_T4463209
Inhousedata_T4465935
Inhousedata_T4469479
Inhousedata_T4471438
NPJ_sample1
NPJ_sample2
NPJ_sample3
NPJ_sample4
NPJ_sample5
NPJ_sample6
NPJ_sample7
CAF
TAM
Treg
-0.6
0.6
Pearson r

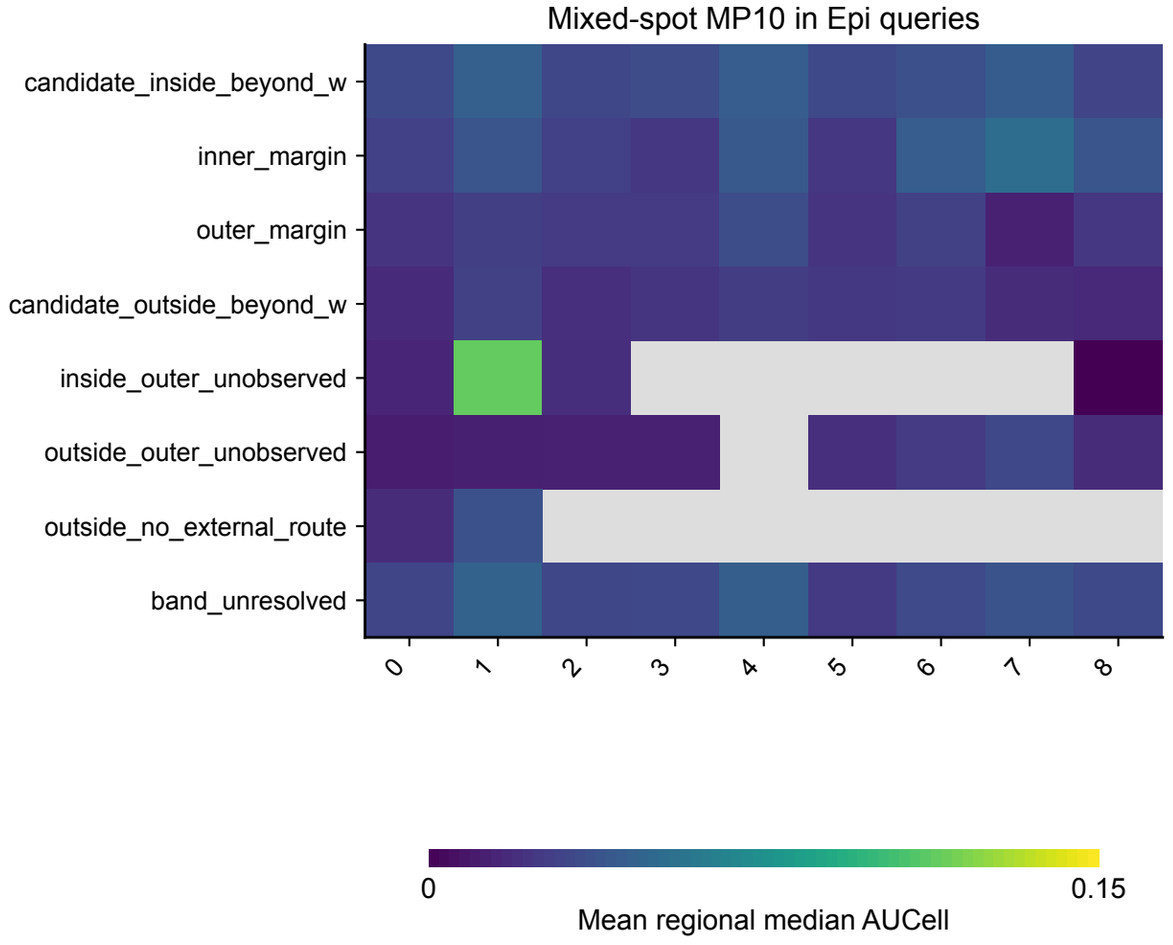

tumor 3 | MP10–distance effects
Within-section Pearson r • positive = higher score farther from target • grey = unavailable
Paired raw
Adjusted
Model spot
Model Epi
GSE319536_BC15
GSE319536_BC16
GSE319536_BC17
GSE319536_BC18
GSE319536_BC19
GSE319536_BC21
Inhousedata_T4384654
Inhousedata_T4398343
Inhousedata_T4413900
Inhousedata_T4463209
Inhousedata_T4465935
Inhousedata_T4469479
Inhousedata_T4471438
NPJ_sample1
NPJ_sample2
NPJ_sample3
NPJ_sample4
NPJ_sample5
NPJ_sample6
NPJ_sample7
CAF
TAM
Treg
-0.6
0.6
Pearson r


100%
75%
50%
25%
0%
Mean reference fraction
inside
outside
v3.4 candidate location
Lineage
Epithelial
Fibroblast
Myeloid
Endothelial
Mural
B
Plasma
T_NK
Mast

Tumor study sections | R12 by v3.4 band

Mean across sections within dataset, then equal datasets • regional MP10 medians require ≥5 Epi queries • grey = NA

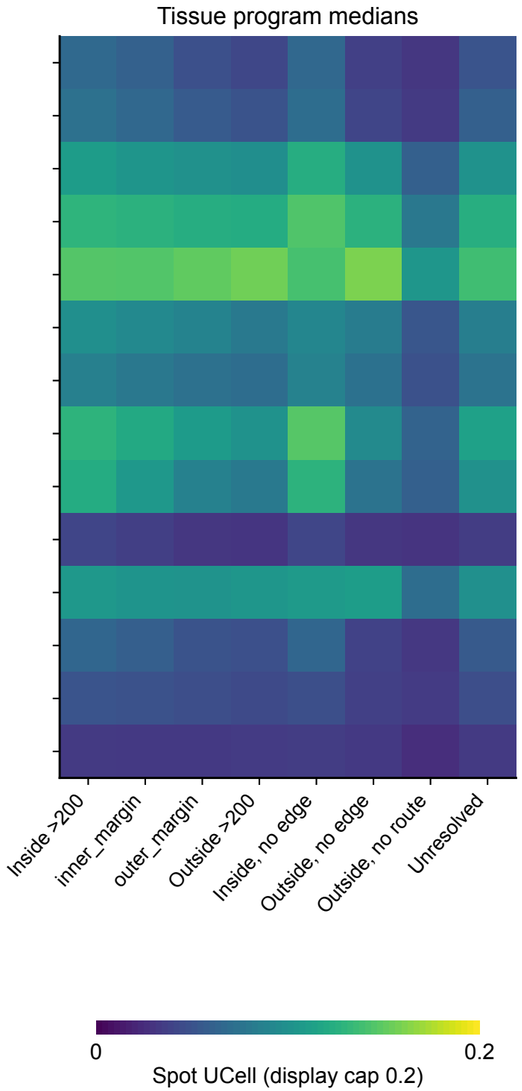


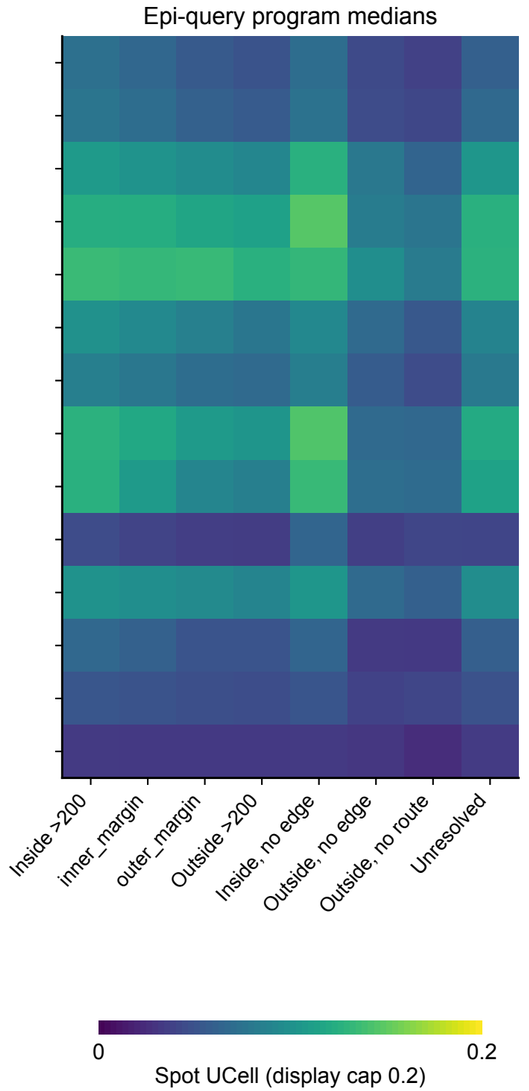


Tumor study sections | Prespecified programs

Distance effects: dataset-equal summary of section medians • region maps: equal dataset means of section medians

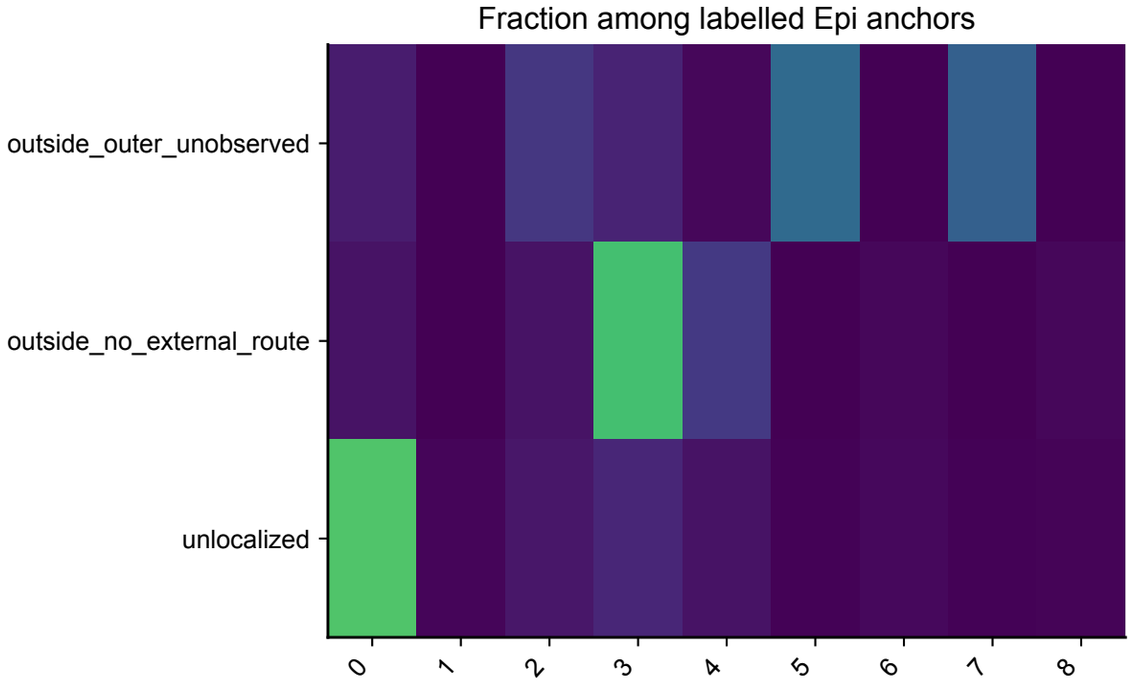


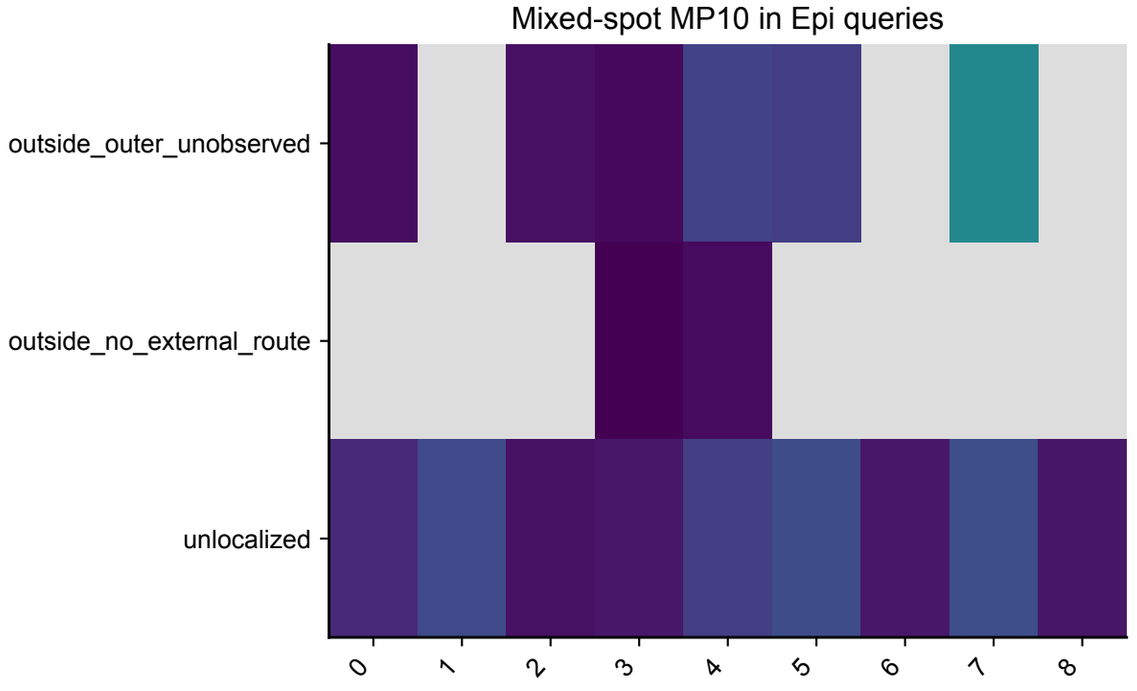




Normal study sections | R12 by v3.4 band

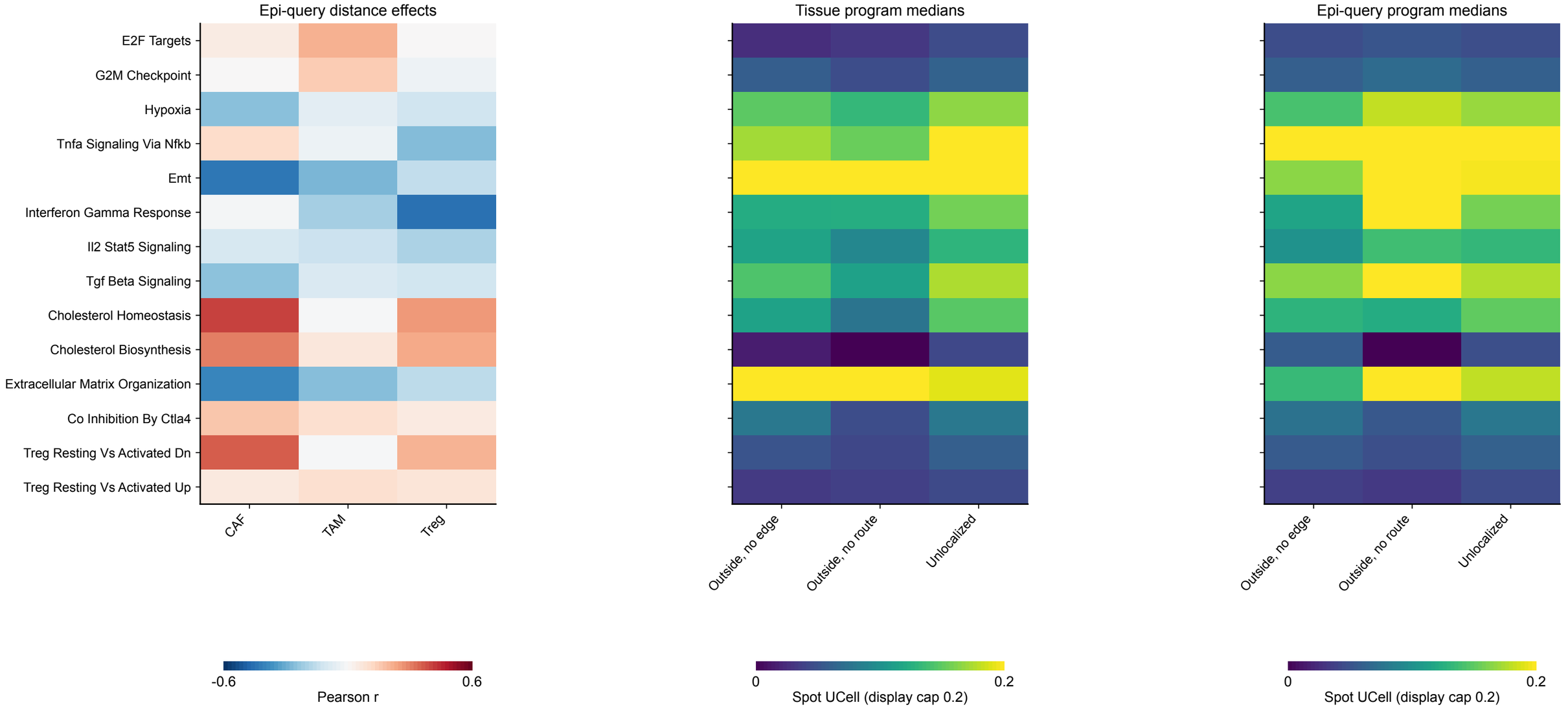
Mean across sections within dataset, then equal datasets • regional MP10 medians require ≥5 Epi queries • grey = NA





Normal study sections | Prespecified programs

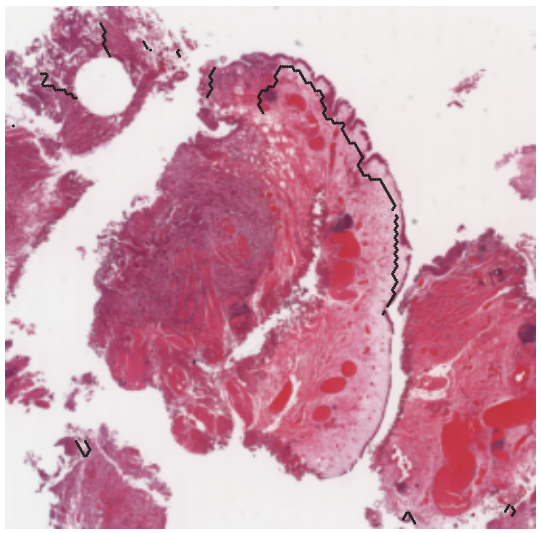
Distance effects: dataset-equal summary of section medians • region maps: equal dataset means of section medians



GSE319536_BC01 | Tissue context

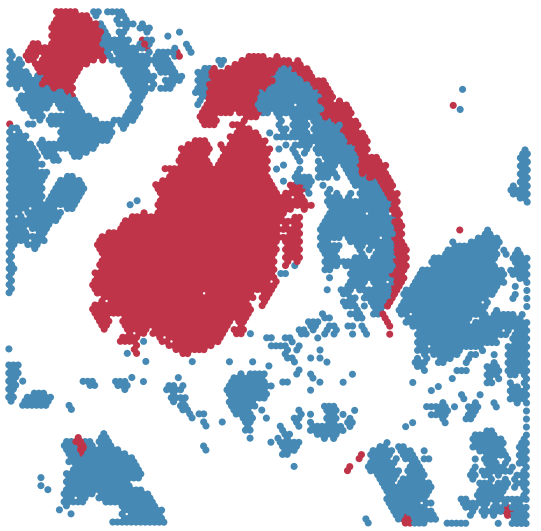
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



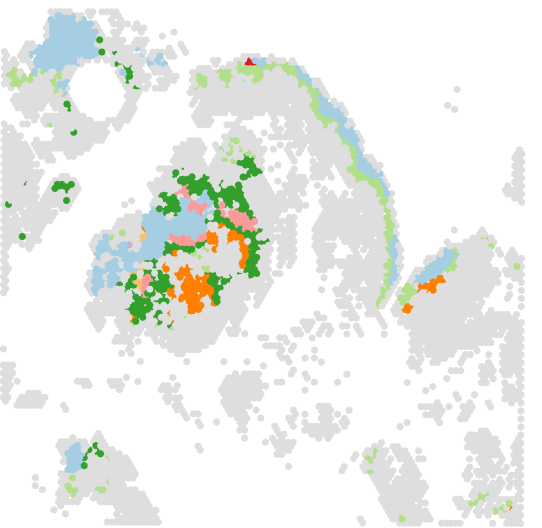
Black line: observed candidate outer edge

Final candidate location



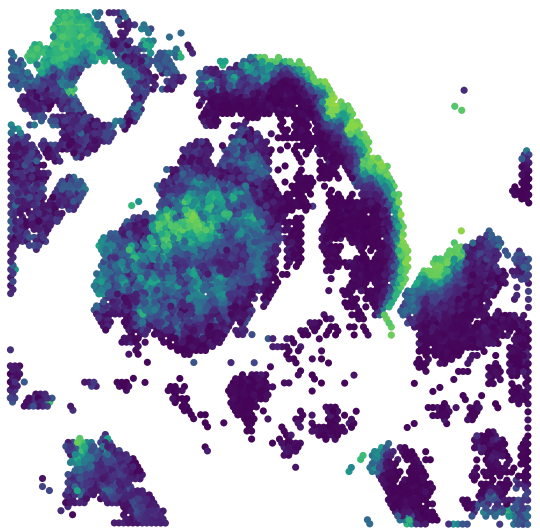
inside outside

Accepted R12 (original 0–8)



0 3 5 7 8
2 4 6

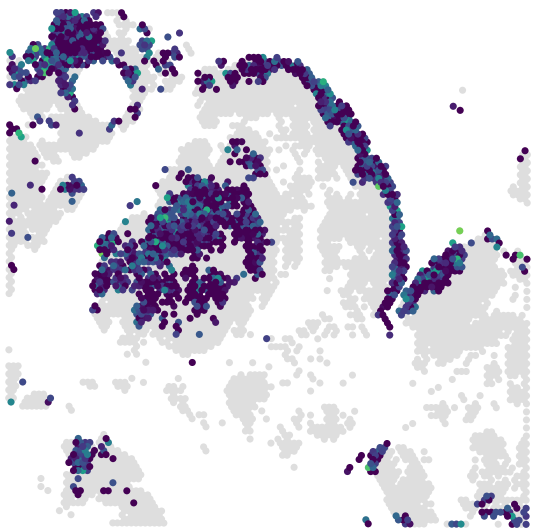
Epithelial support



0 1

Fraction of all 30

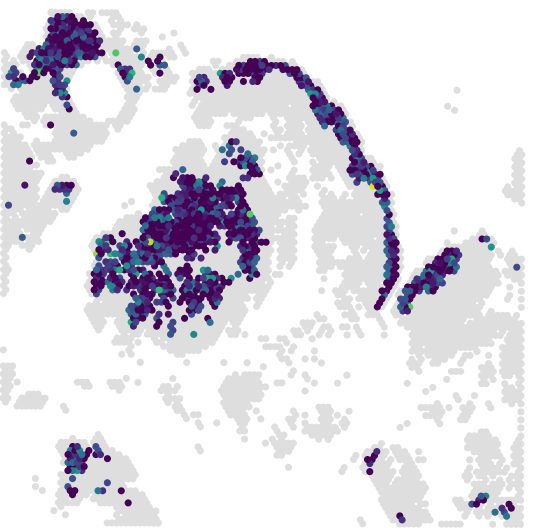
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



0 0.14

Matched-model AUCell

CanDis_2026_Sample9 | Tissue context

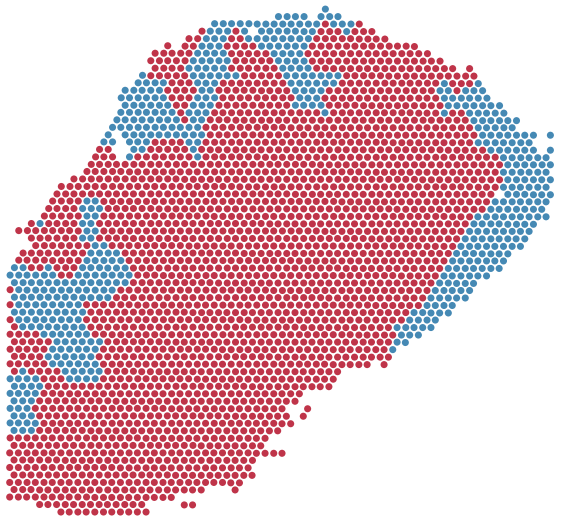
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



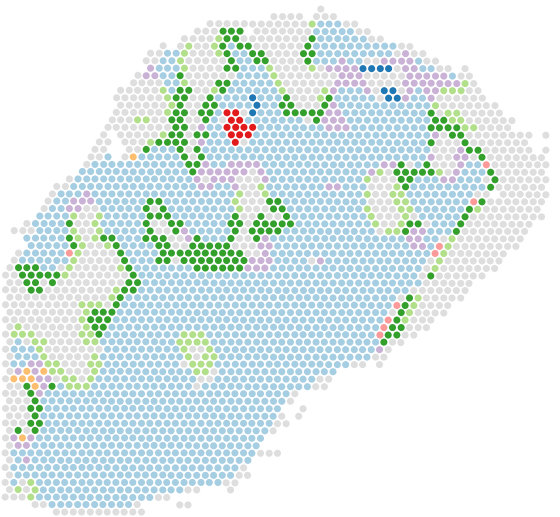
Black line: observed candidate outer edge

Final candidate location



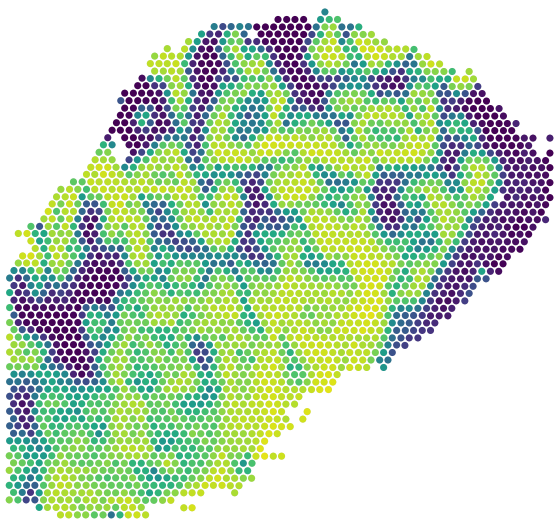
inside outside

Accepted R12 (original 0–8)



0 2 4 7 8
1 3 6

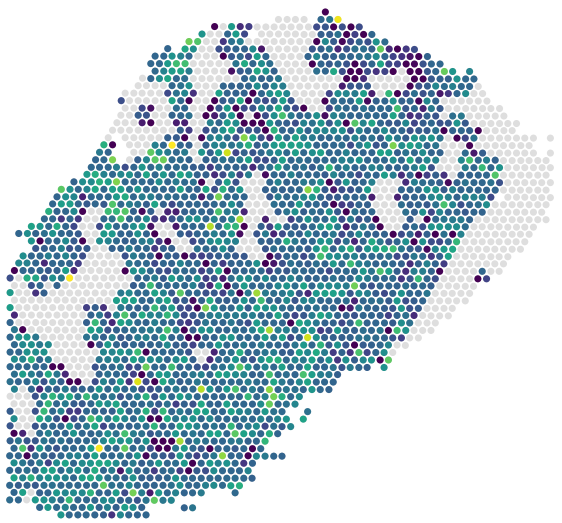
Epithelial support



0 1

Fraction of all 30

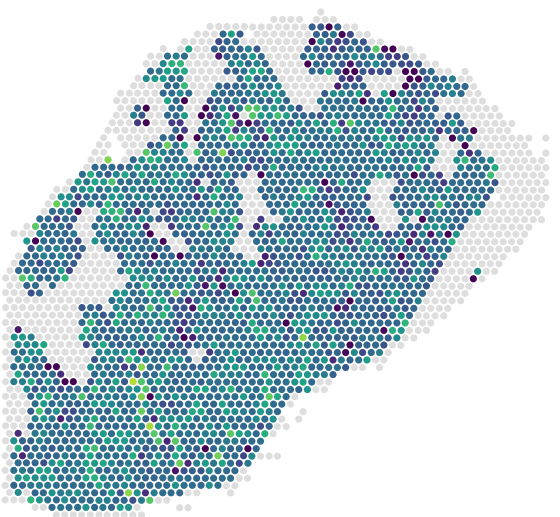
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



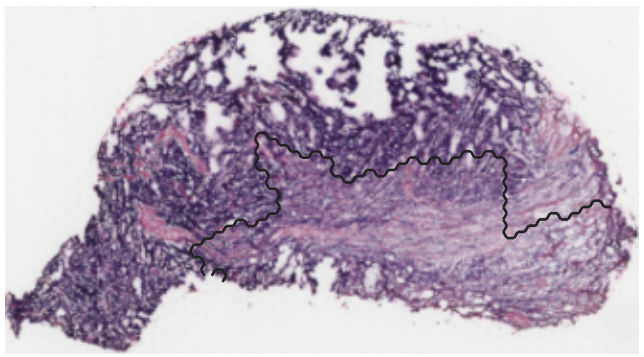
0 0.14

Matched-model AUCell

GSE171351_B1 | Tissue context

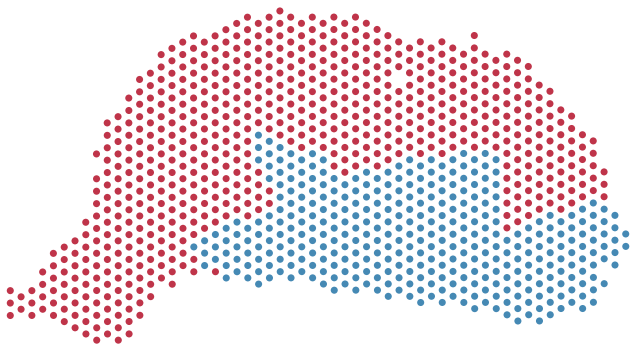
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



Black line: observed candidate outer edge

Final candidate location



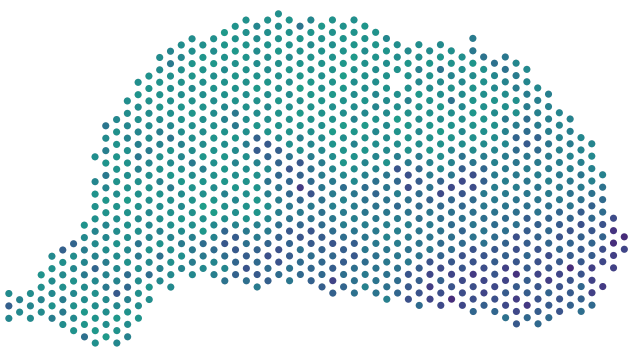
inside outside

Accepted R12 (original 0–8)



1 2 3 4 6

Epithelial support



0 1

Fraction of all 30

MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



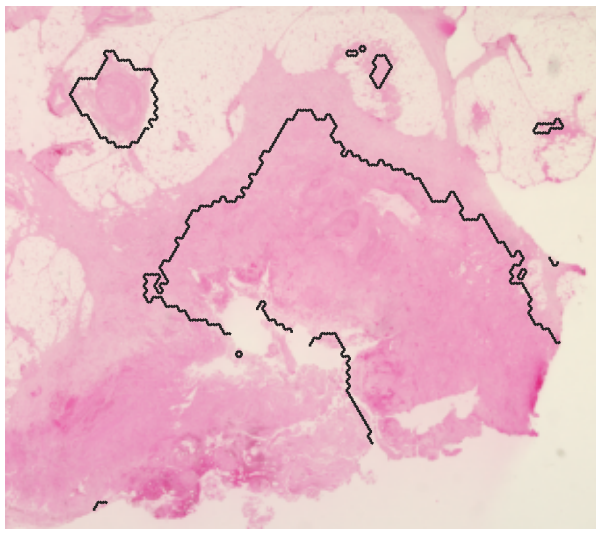
0 0.14

Matched-model AUCell

GSE238145_B14 | Tissue context

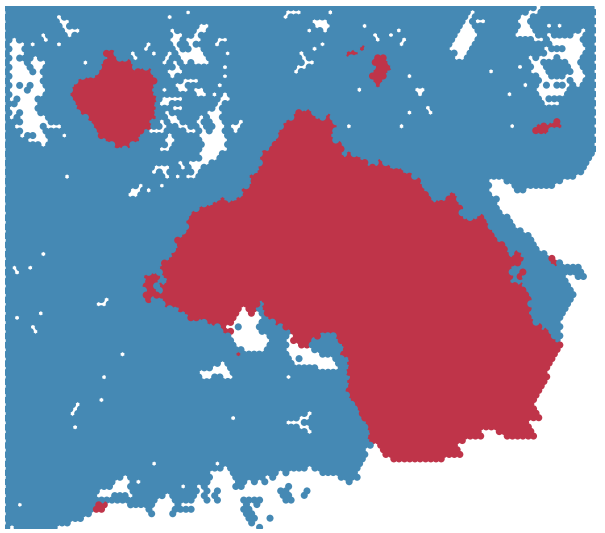
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



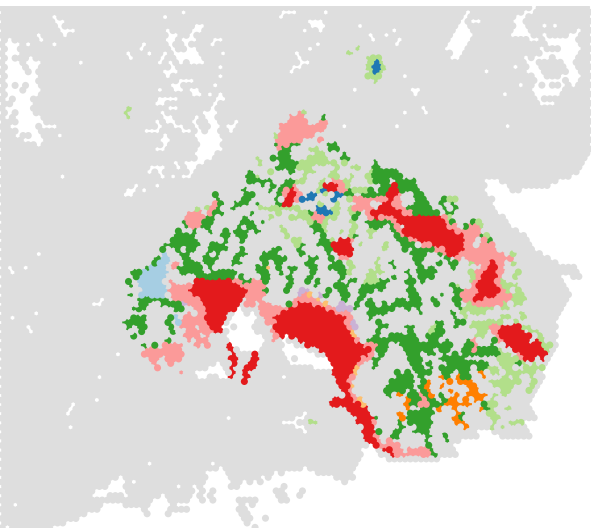
Black line: observed candidate outer edge

Final candidate location



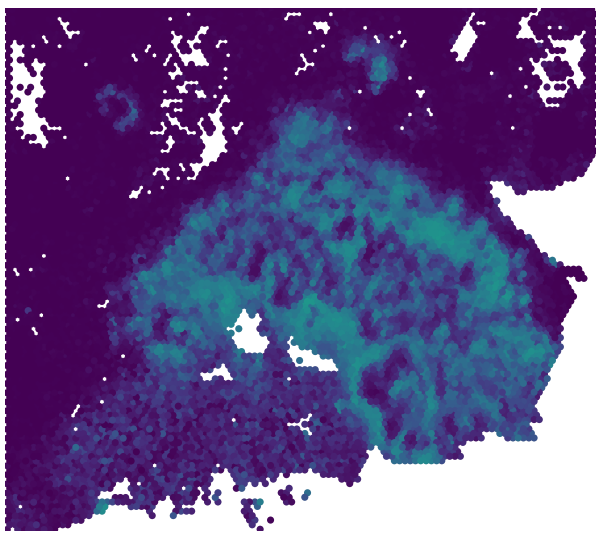
inside outside

Accepted R12 (original 0–8)



0 2 4 6 8
1 3 5 7

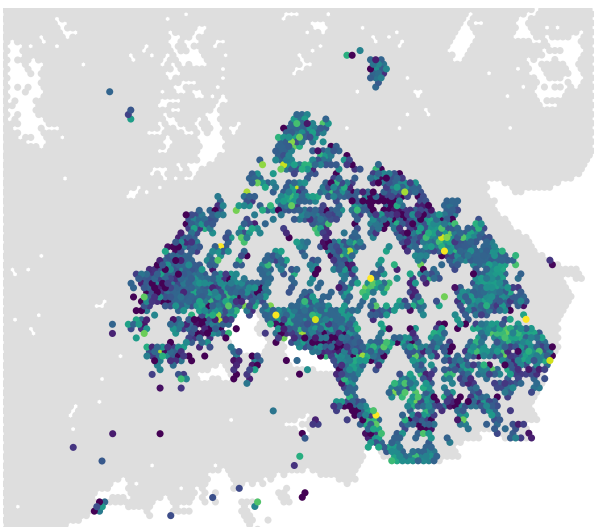
Epithelial support



0 1

Fraction of all 30

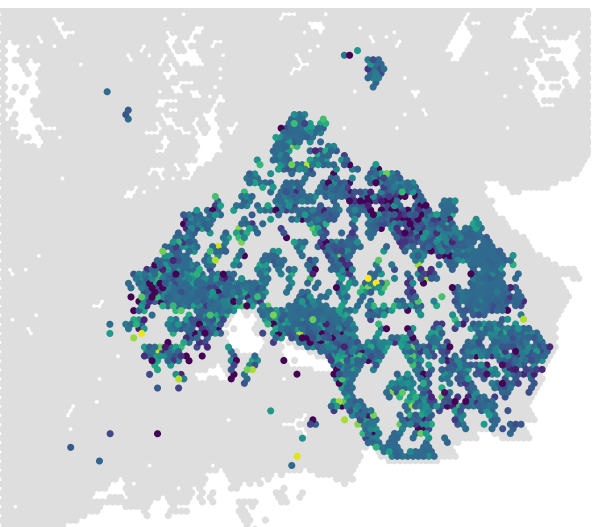
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



0 0.14

Matched-model AUCell

GSE285715_T2 | Tissue context

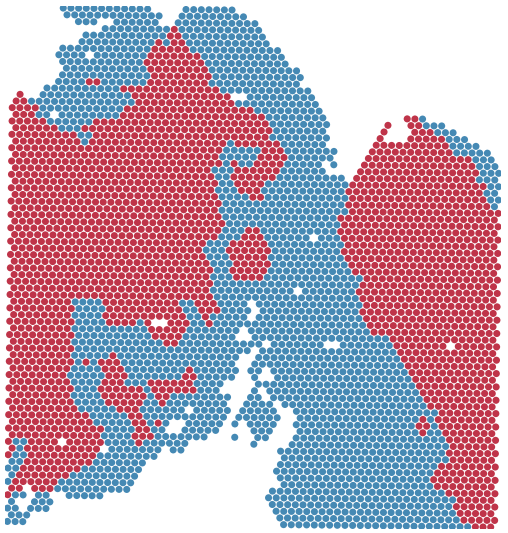
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



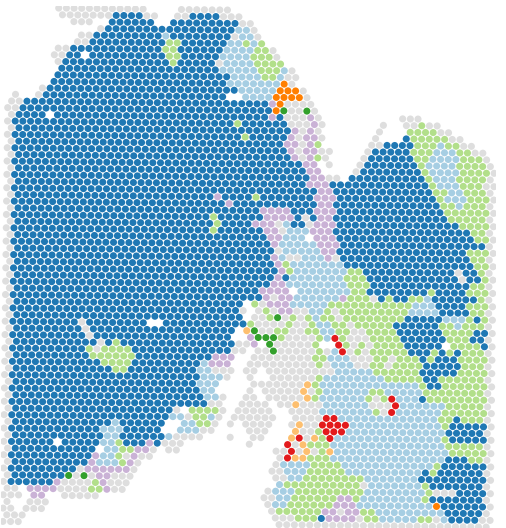
Black line: observed candidate outer edge

Final candidate location



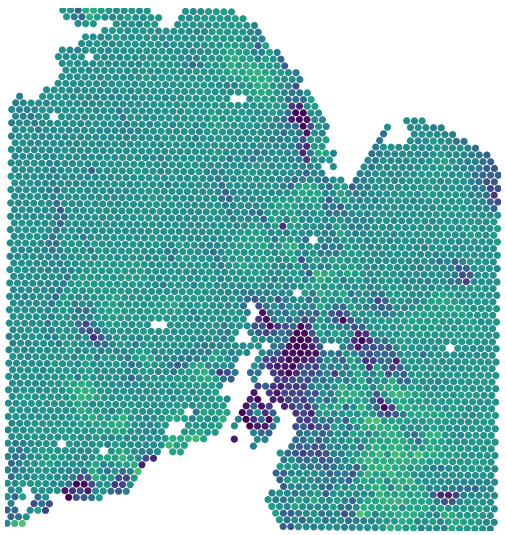
inside outside

Accepted R12 (original 0–8)



0 2 4 7 8
1 3 5

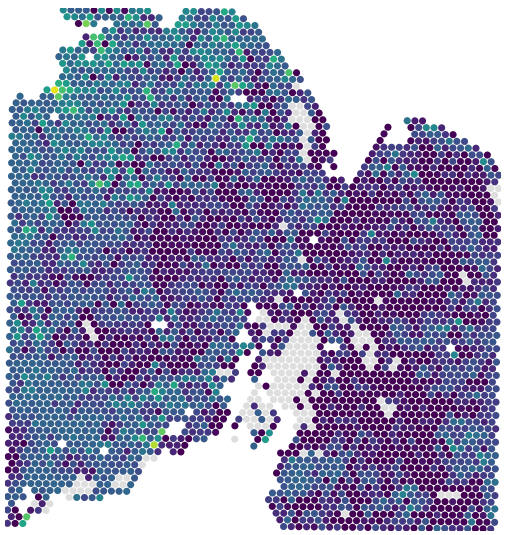
Epithelial support



0 1

Fraction of all 30

MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



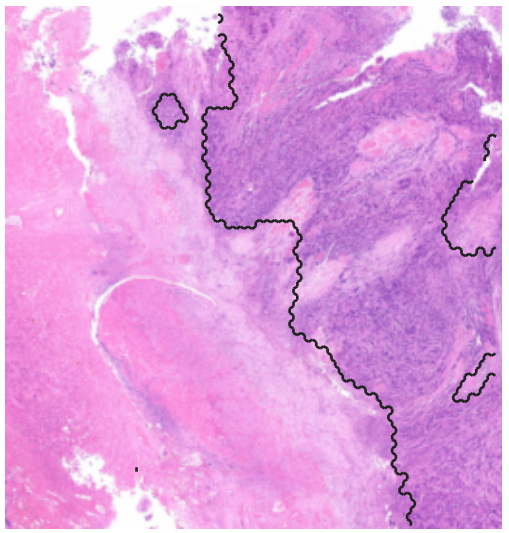
0 0.14

Matched-model AUCell

Inhousedata_T4469479 | Tissue context

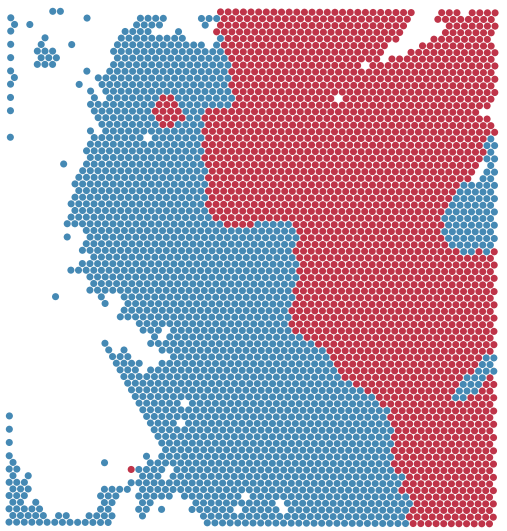
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



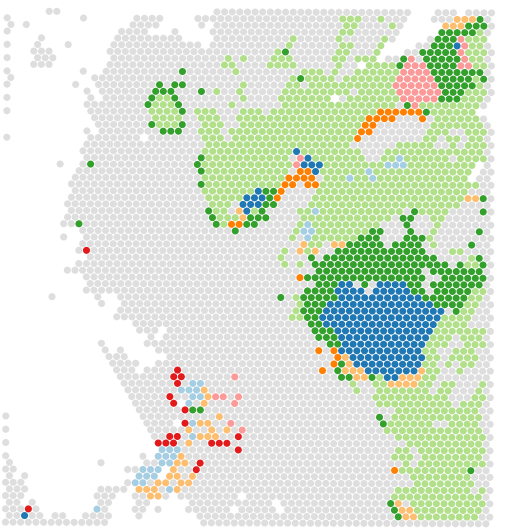
Black line: observed candidate outer edge

Final candidate location



inside outside

Accepted R12 (original 0–8)



0 2 4 6 7
1 3 5

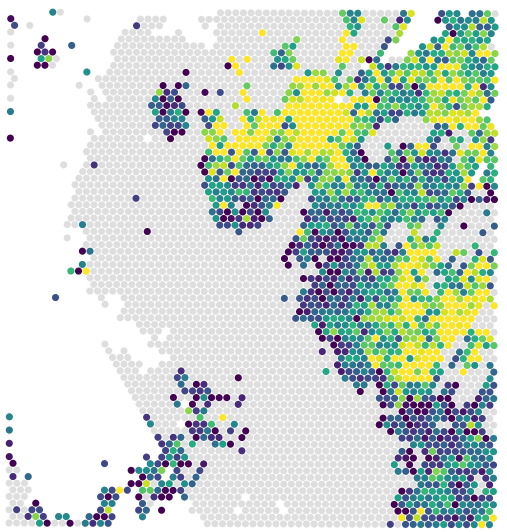
Epithelial support



0 1

Fraction of all 30

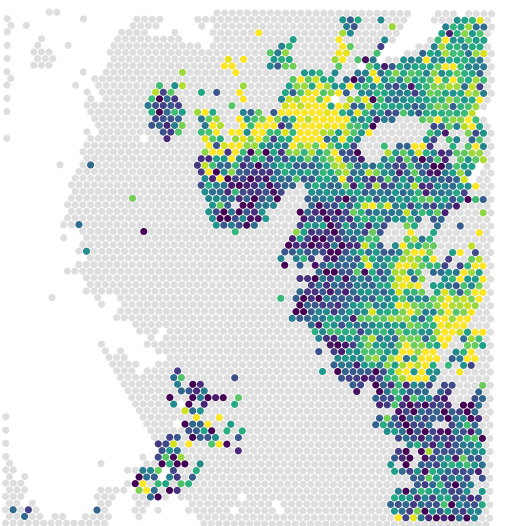
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



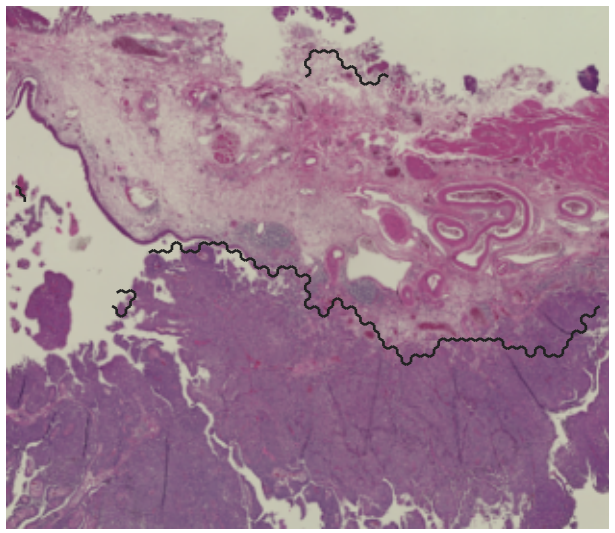
0 0.14

Matched-model AUCell

NPJ_sample6 | Tissue context

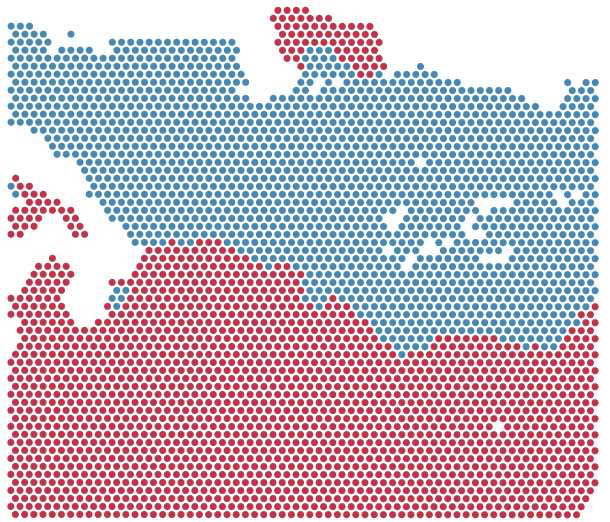
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



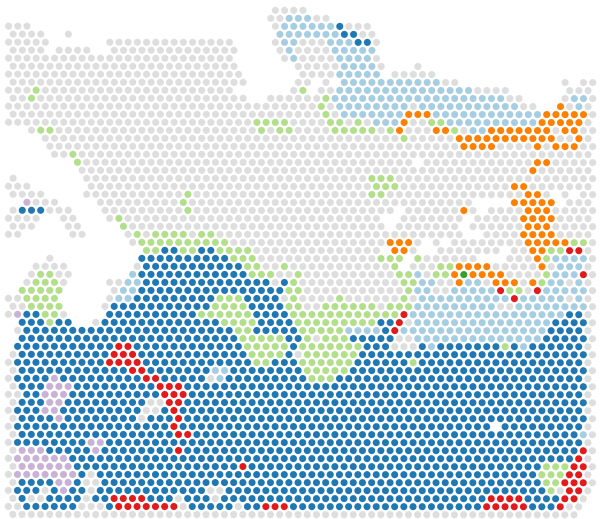
Black line: observed candidate outer edge

Final candidate location

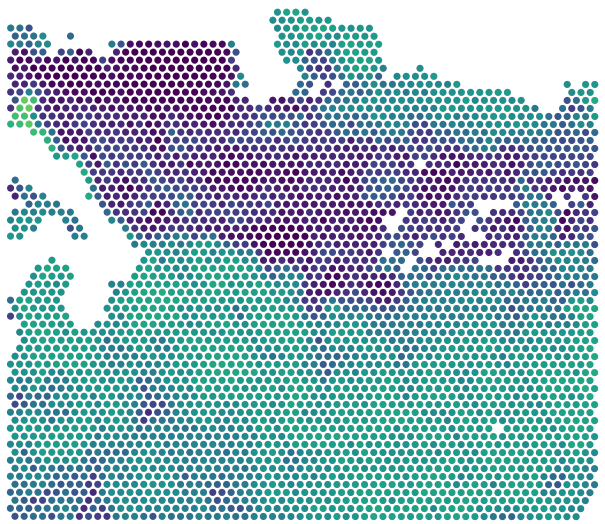




Accepted R12 (original 0–8)



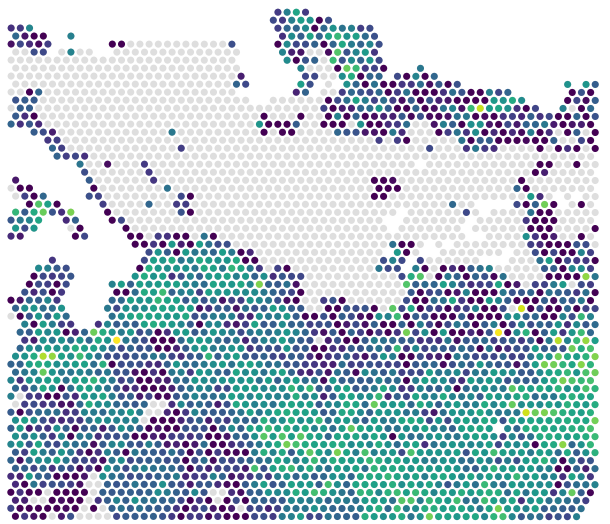
Epithelial support





Fraction of all 30

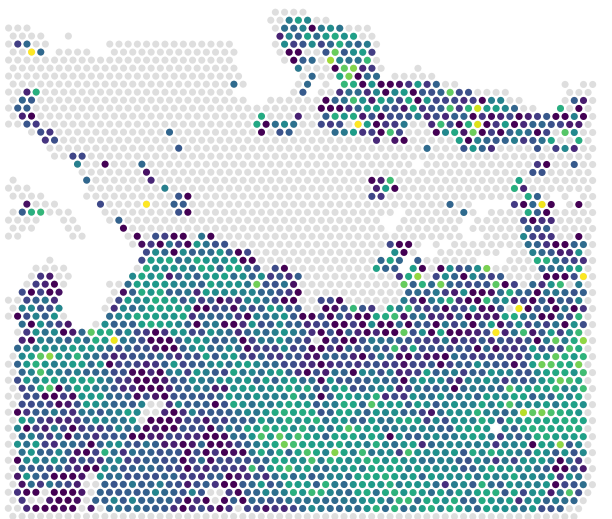
MP10 in Epi queries





Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)





Matched-model AUCell